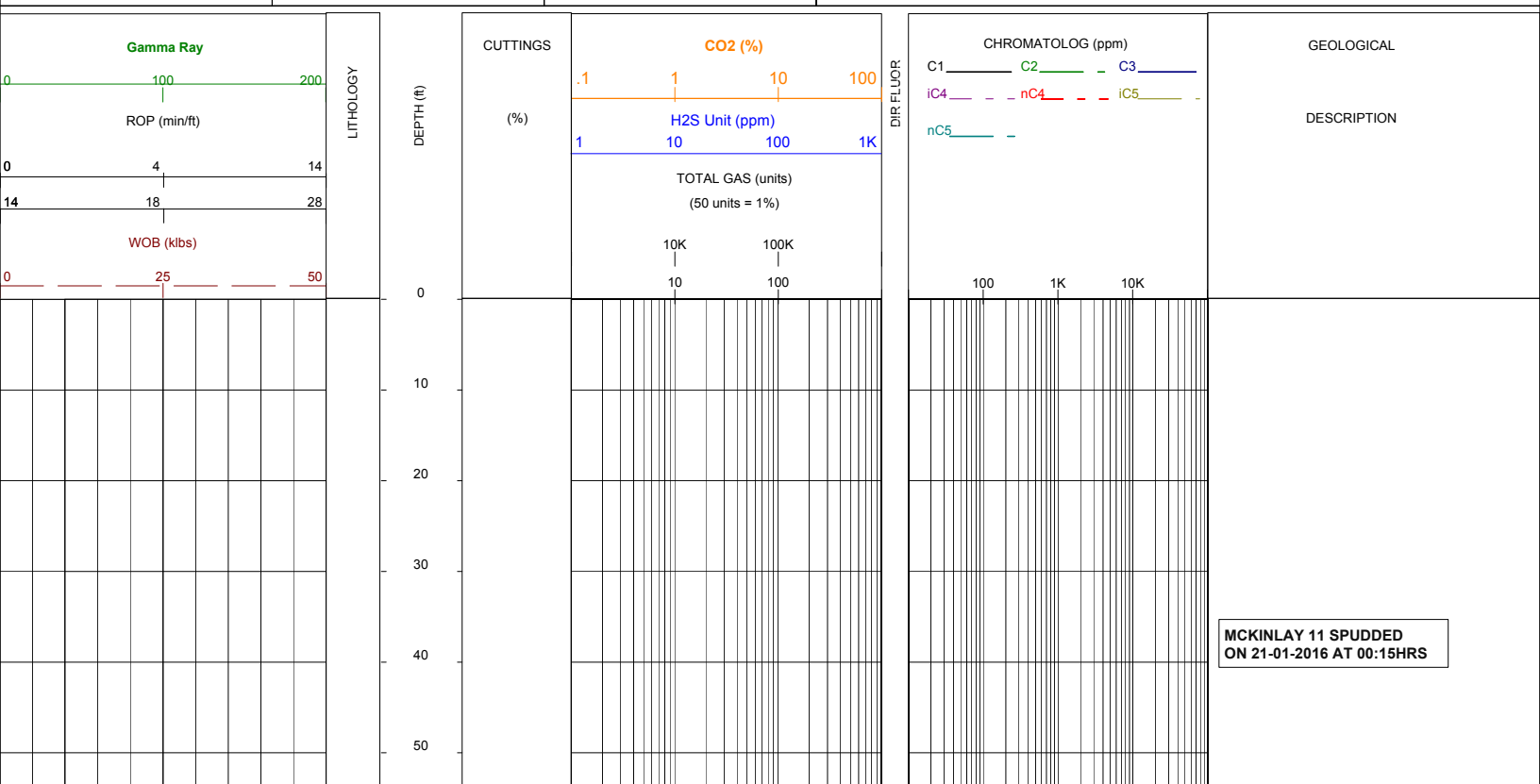
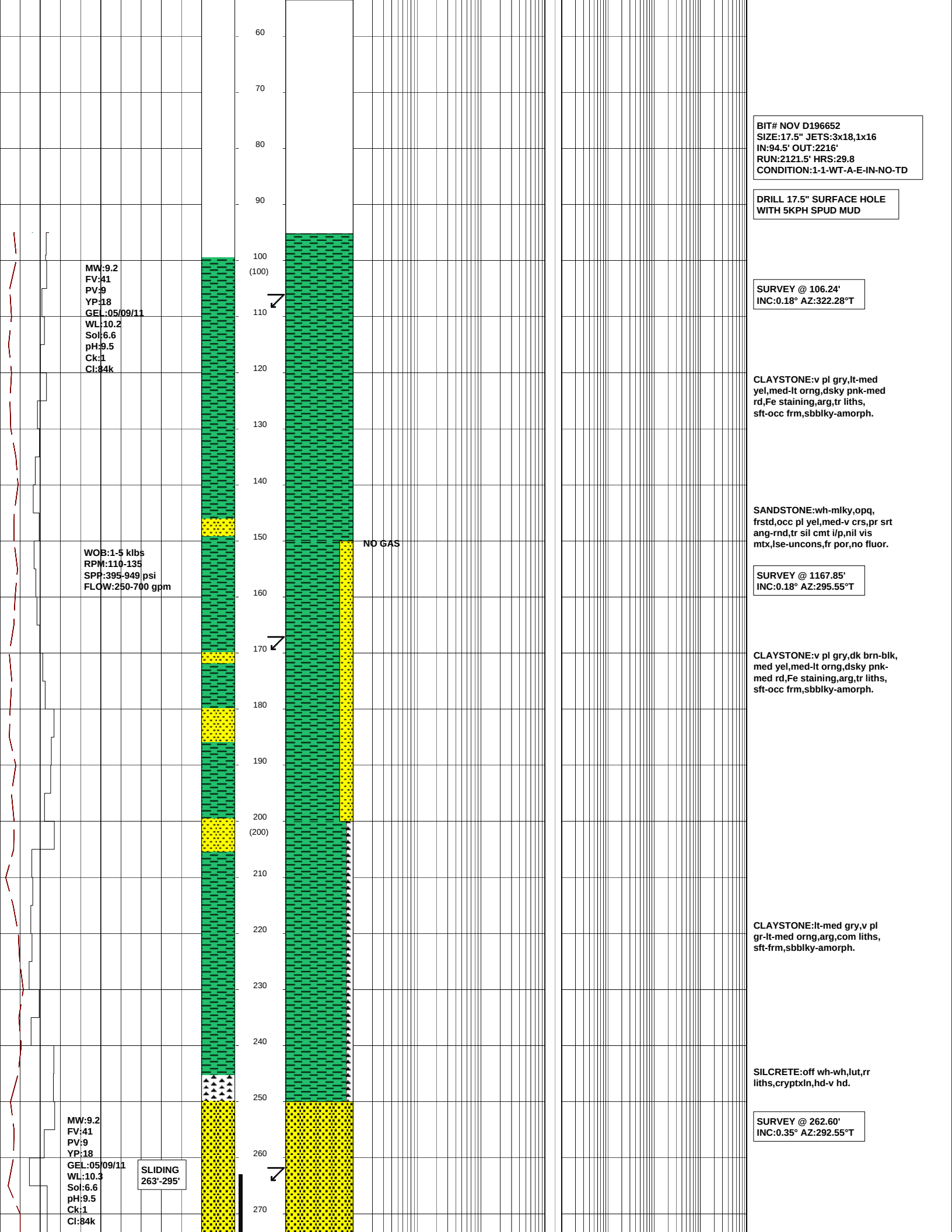
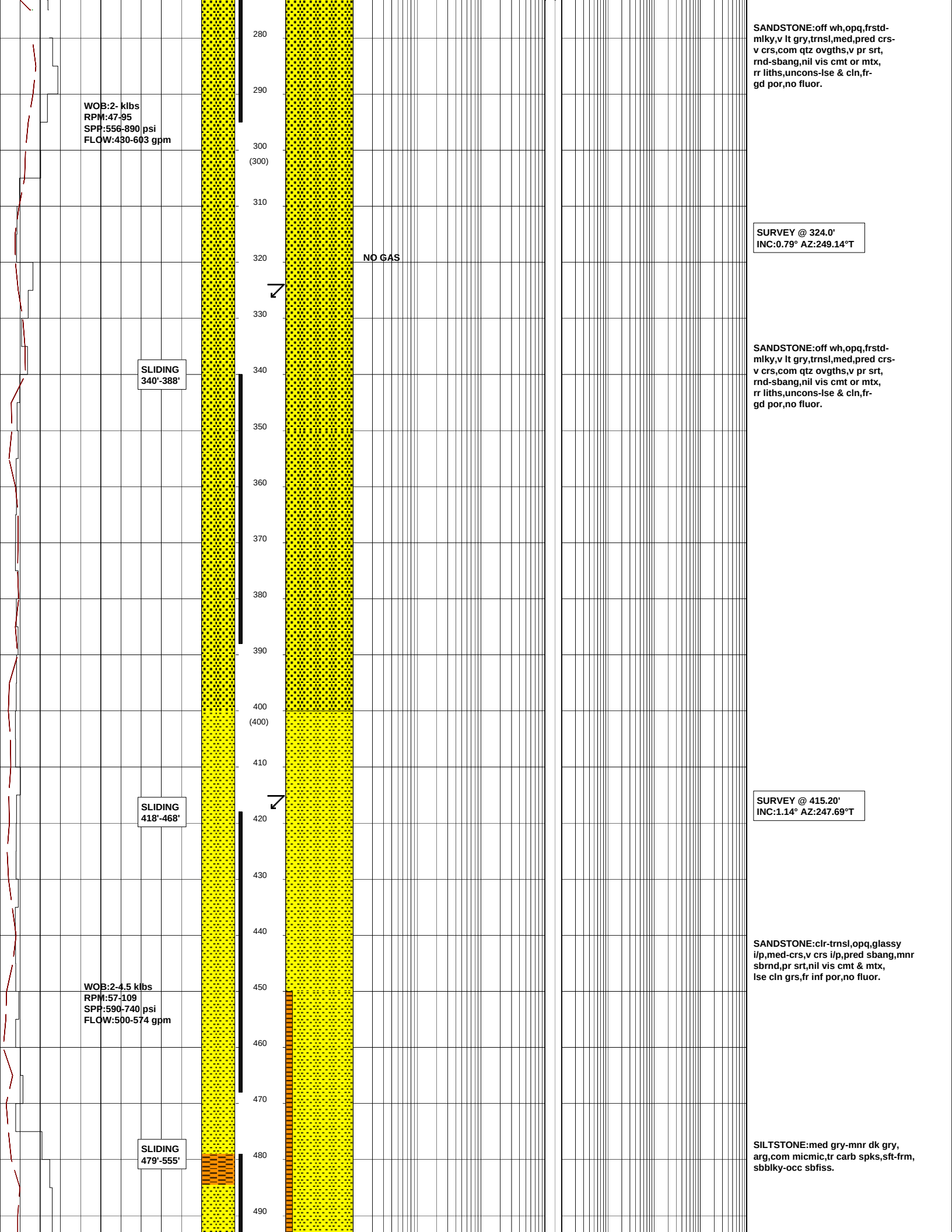


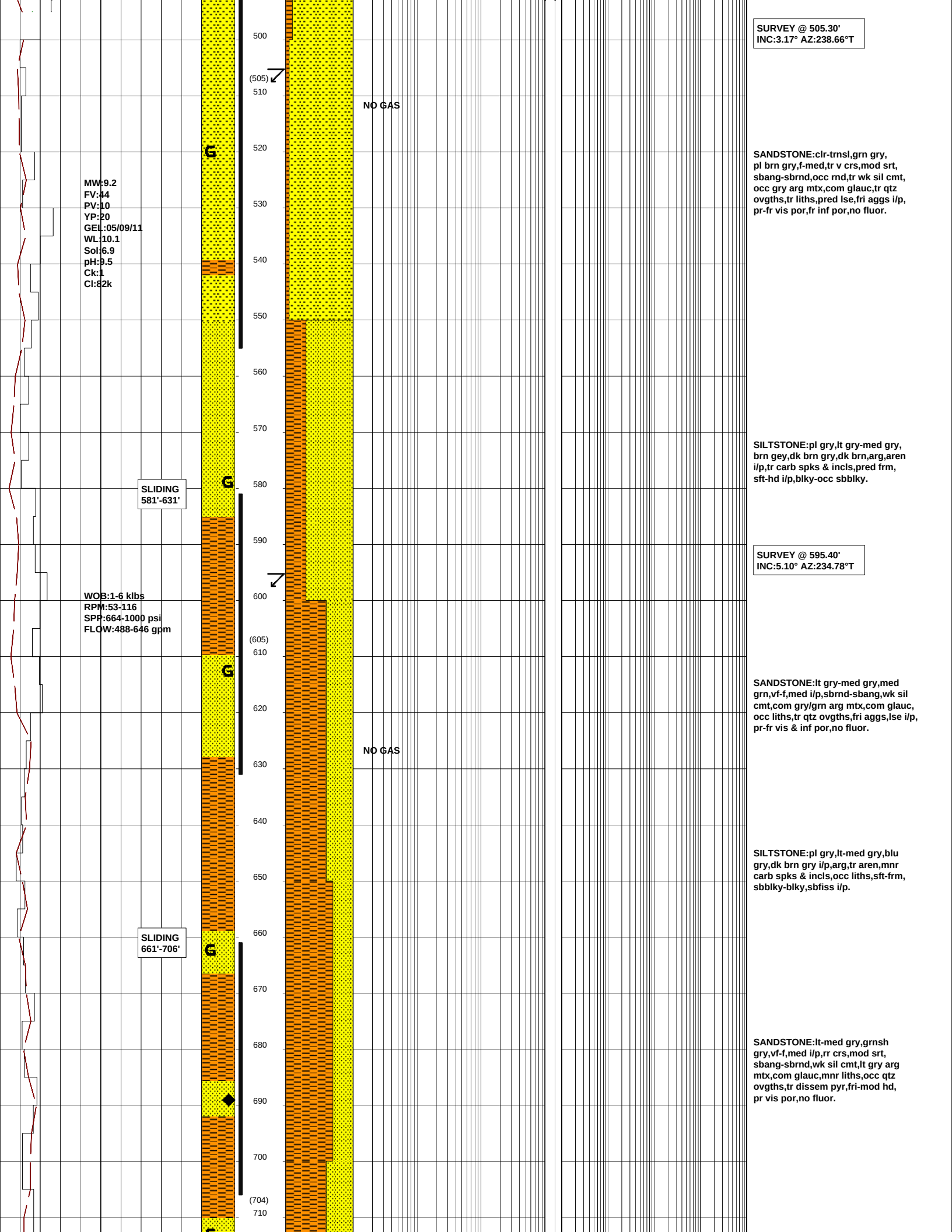
Field : MCKINLAY	Rotary Table : 149.3'	Rig : ENSIGN 971	Open Hole:	Cased Hole:	Loggers :	K.BRZOZOWSKA
Permit : PPL234	Ground Level : 125.9'	Spud Date : 21-01-2016	17.50" 2216'	13.375" 2206'		JASON MADIGAN
State : SOUTH AUSTRALIA	GDA 94 Co-ordinates :	TD Date : 02-02-2016	12.25" 5124'	9.625" 5114'		VAHID JALEH
Country : AUSTRALIA	Lat. : 28°28'57.49"S	Total Depth : 6786'MD 4068'TVD	8.5" 6786'	5.5" 6776'		ALEX MCINNES
Scale : 1/240 5":100"	Long. : 140°28'39.28"E	Final Status : CASED & SUSPENDED				JESS NUTTALL

LITHOLOGY	ACCESSORIES	DRILLING DATA	ABBREVIATIONS
 Coarse sandstone  Medium sandstone  Fine sandstone  VF Sandstone  Siltstone  Carb Siltstone  Metamorphic rock  Claystone  Limestone  Coal-lignite	 Pyrite  Siderite  Glauconite  Iron  Mica  Chert  Calcareous  Dolomitic  Carbonaceous  Fossils	 Casing shoe  Bit Trip  Wiper Trip  Core  Dst  Survey MUD DATA MW - Mud Weight (lb/gal) FV - Funnel Viscosity (s/qt) PV - Plastic Viscosity (cps) YP - Yield Point (lb/100ftsq) Gel - Gel Strength (10sec) WL - Water Loss (cc/30min) pH - Acidity / Alkalinity Ck - Cake (32nd/inch) Sol - Solids (% vol) Cl - Chlorides (mg/l)	ABBREVIATIONS BOPD - Barrels of Oil Per Day BWPD - Barrels of Water Per Day CG - Connection Gas CO - Circulate Out COND - Condensate c/c - Crush Cut DST - Drill Stem Test FLOW - Flow Rate (gal/min) GCM - Gas Cut Mud GCW - Gas Cut Water GTS - Gas To Surface INJ - Injection of Mist (bbls/hr) LCM - Lost Circulation Material MMCFD - Million Cubic Feet / Day NGTS - No Gas To Surface NOTS - No Oil To Surface NR - No Returns OCM - Oil Cut Mud OG - Over Gauge OH - Open Hole OTS - Oil To Surface Q - Flow Rate REC - Recovery Rmf - Resistivity mud filtrate ROP - Rate Of Penetration RPM - Revolutions Per Minute RTSTM - Rate Too Small To Measure Rw - Resistivity water r/r - Ring Residue SCFM - Standard Cubic Ft/Min (air) SGCM - Slightly Gas Cut Mud SPM - Strokes Per Minute SPP - Stand Pipe Pressure SWC - Side-Wall Core TG - Trip Gas WOB - Weight On Bit









SURVEY @ 505.30'
INC: 3.17° AZ: 238.66°T

MW: 9.2
FV: 44
PV: 10
YP: 20
GEL: 05/09/11
WL: 10.1
Sol: 6.9
pH: 9.5
Ck: 1
Cl: 82k

SLIDING
581'-631'

WOB: 1-6 klbs
RPM: 53-116
SPR: 664-1000 psi
FLOW: 488-646 gpm

SLIDING
661'-706'

SANDSTONE: clr-trns, grn gry, pl brn gry, f-med, tr v crs, mod srt, sbang-sbrnd, occ rnd, tr wk sil cmt, occ gry arg mtx, com glauc, tr qtz ovghts, tr liths, pred lse, fri aggs i/p, pr-fr vis por, fr inf por, no fluor.

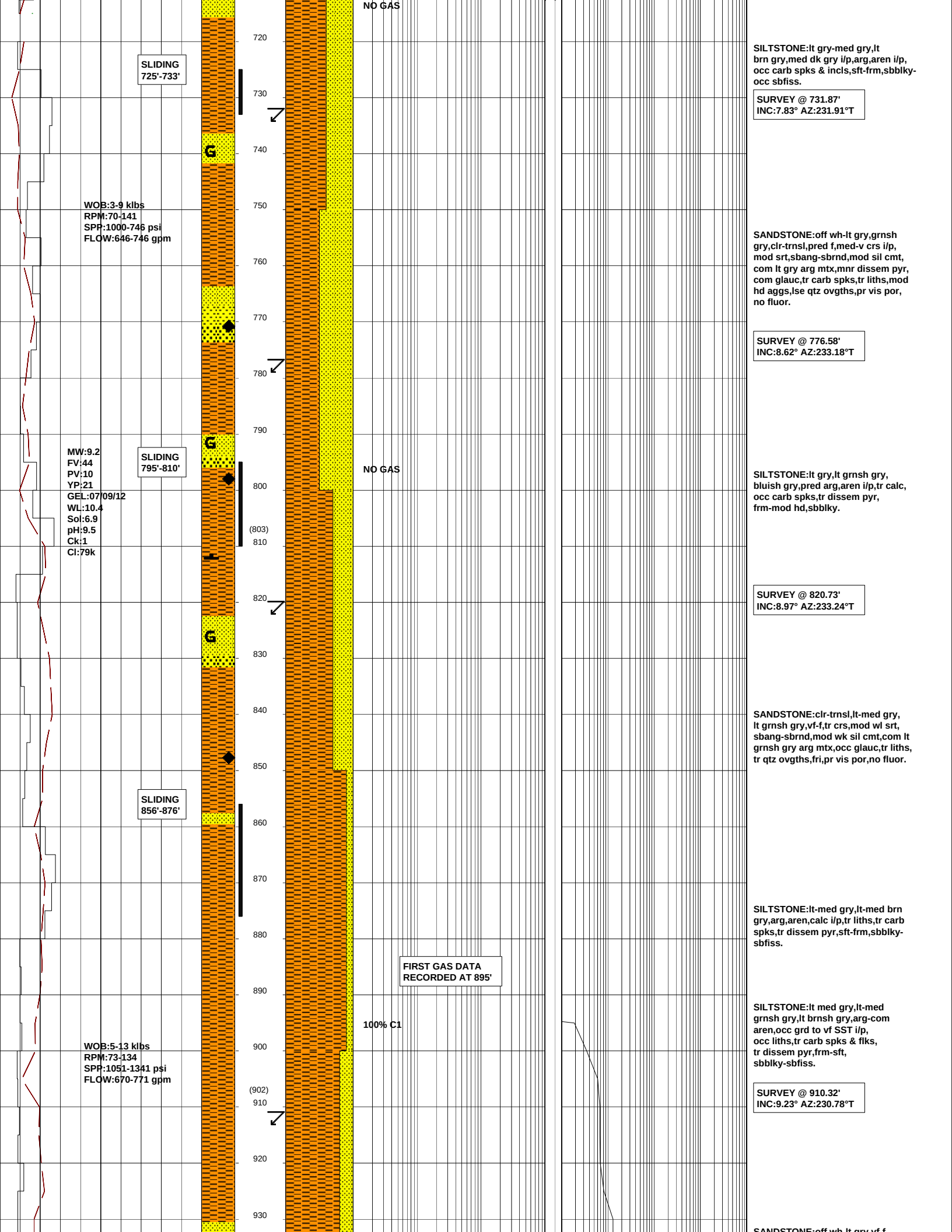
SILTSTONE: pl gry, lt gry-med gry, brn gey, dk brn gry, dk brn, arg, aren i/p, tr carb spks & inclcs, pred frm, sft-hd i/p, blk-occ sbblky.

SURVEY @ 595.40'
INC: 5.10° AZ: 234.78°T

SANDSTONE: lt gry-med gry, med grn, vf-f, med i/p, sbrnd-sbang, wk sil cmt, com gry/grn arg mtx, com glauc, occ liths, tr qtz ovghts, fri aggs, lse i/p, pr-fr vis & inf por, no fluor.

SILTSTONE: pl gry, lt-med gry, blu gry, dk brn gry i/p, arg, tr aren, mnr carb spks & inclcs, occ liths, sft-frm, sbblky-blky, sbfiss i/p.

SANDSTONE: lt-med gry, grnsh gry, vf-f, med i/p, rr crs, mod srt, sbang-sbrnd, wk sil cmt, lt gry arg mtx, com glauc, mnr liths, occ qtz ovghts, tr disse pyr, fri-mod hd, pr vis por, no fluor.



SLIDING
725'-733'

WOB:3-9 klbs
RPM:70-141
SPP:1000-746 psi
FLOW:646-746 gpm

MW:9.2
FV:44
PV:10
YP:21
GEL:07/09/12
WL:10.4
Sol:6.9
pH:9.5
Ck:1
CI:79k

SLIDING
795'-810'

SLIDING
856'-876'

WOB:5-13 klbs
RPM:73-134
SPP:1051-1341 psi
FLOW:670-771 gpm

FIRST GAS DATA
RECORDED AT 895'

SILTSTONE:lt gry-med gry,lt
brn gry,med dk gry i/p,arg,aren i/p,
occ carb spks & incls,sft-frm,sbblky-
occ sbfiss.

SURVEY @ 731.87'
INC:7.83° AZ:231.91°T

SANDSTONE:off wh-lt gry,grnsh
gry,clr-trnsl,pred f,med-v crs i/p,
mod srt,sbang-sbrnd,mod sil cmt,
com lt gry arg mtx,mnr disse pyr,
com glauc,tr carb spks,tr liths,mod
hd aggs,lse qtz ovghts,pr vis por,
no fluor.

SURVEY @ 776.58'
INC:8.62° AZ:233.18°T

SILTSTONE:lt gry,lt grnsh gry,
bluish gry,pred arg,aren i/p,tr calc,
occ carb spks,tr disse pyr,
frm-mod hd,sbblky.

SURVEY @ 820.73'
INC:8.97° AZ:233.24°T

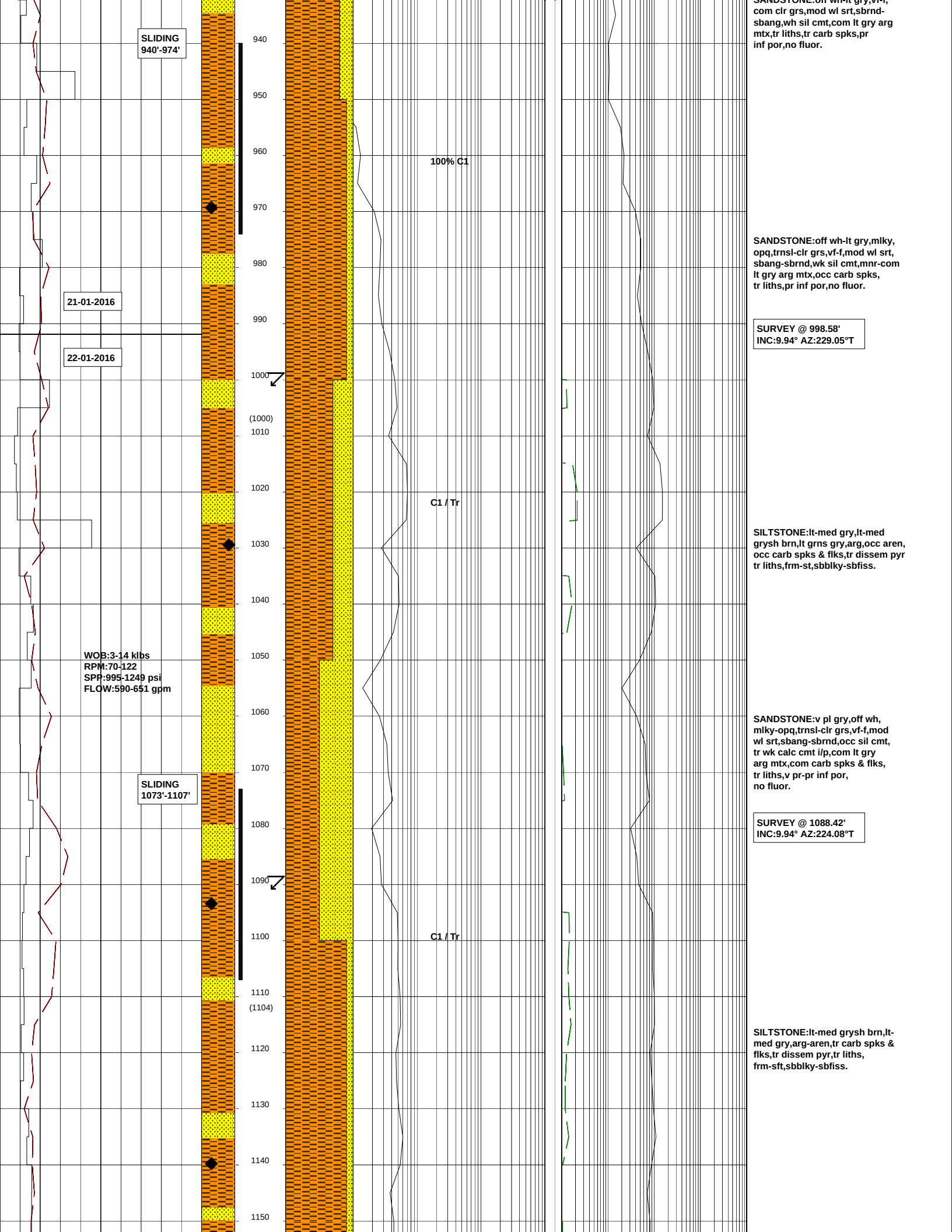
SANDSTONE:clr-trnsl,lt-med gry,
lt grnsh gry,vf-f,tr crs,mod wl srt,
sbang-sbrnd,mod wk sil cmt,com lt
grnsh gry arg mtx,occ glauc,tr liths,
tr qtz ovghts,fri,pr vis por,no fluor.

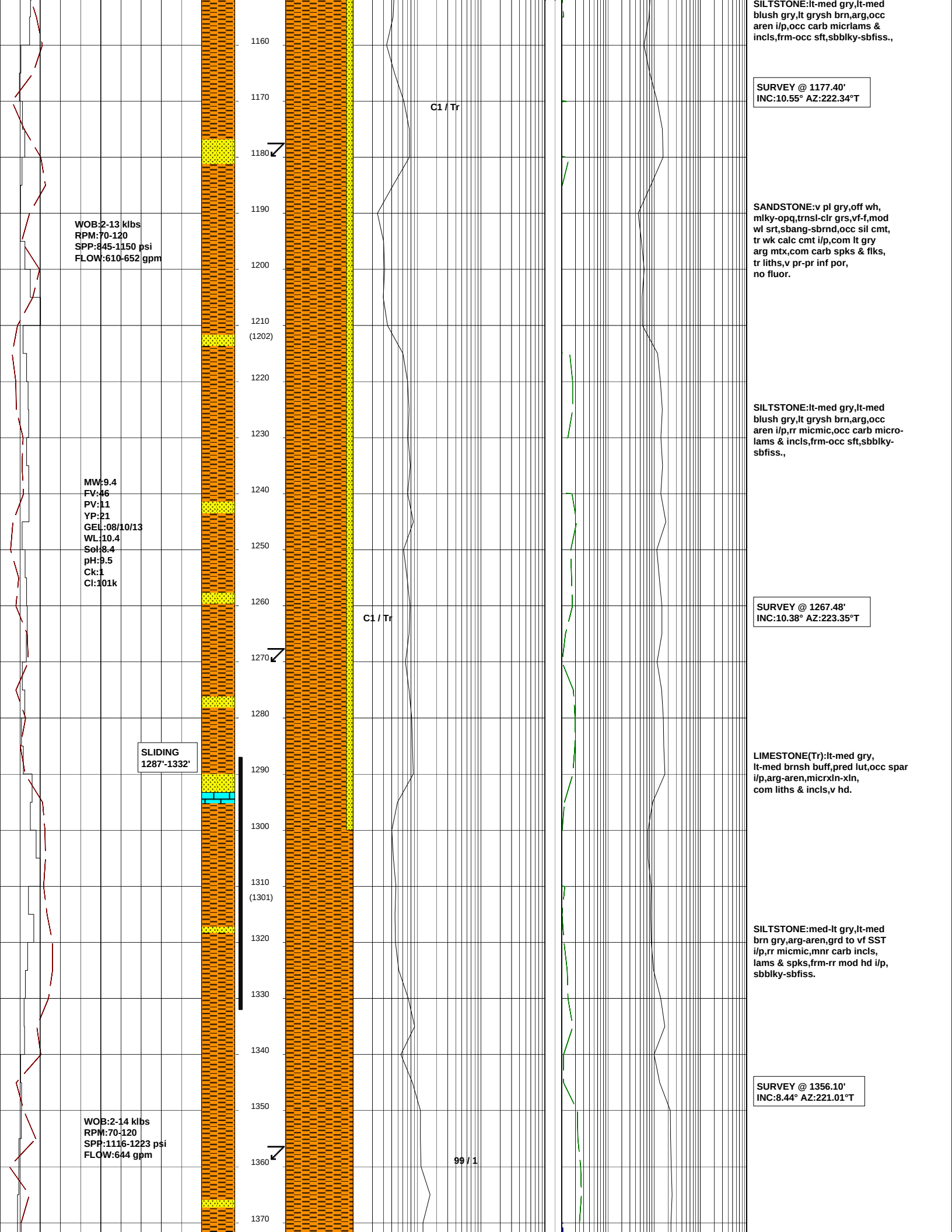
SILTSTONE:lt-med gry,lt-med brn
gry,arg,aren,calc i/p,tr liths,tr carb
spks,tr disse pyr,sft-frm,sbblky-
sbfiss.

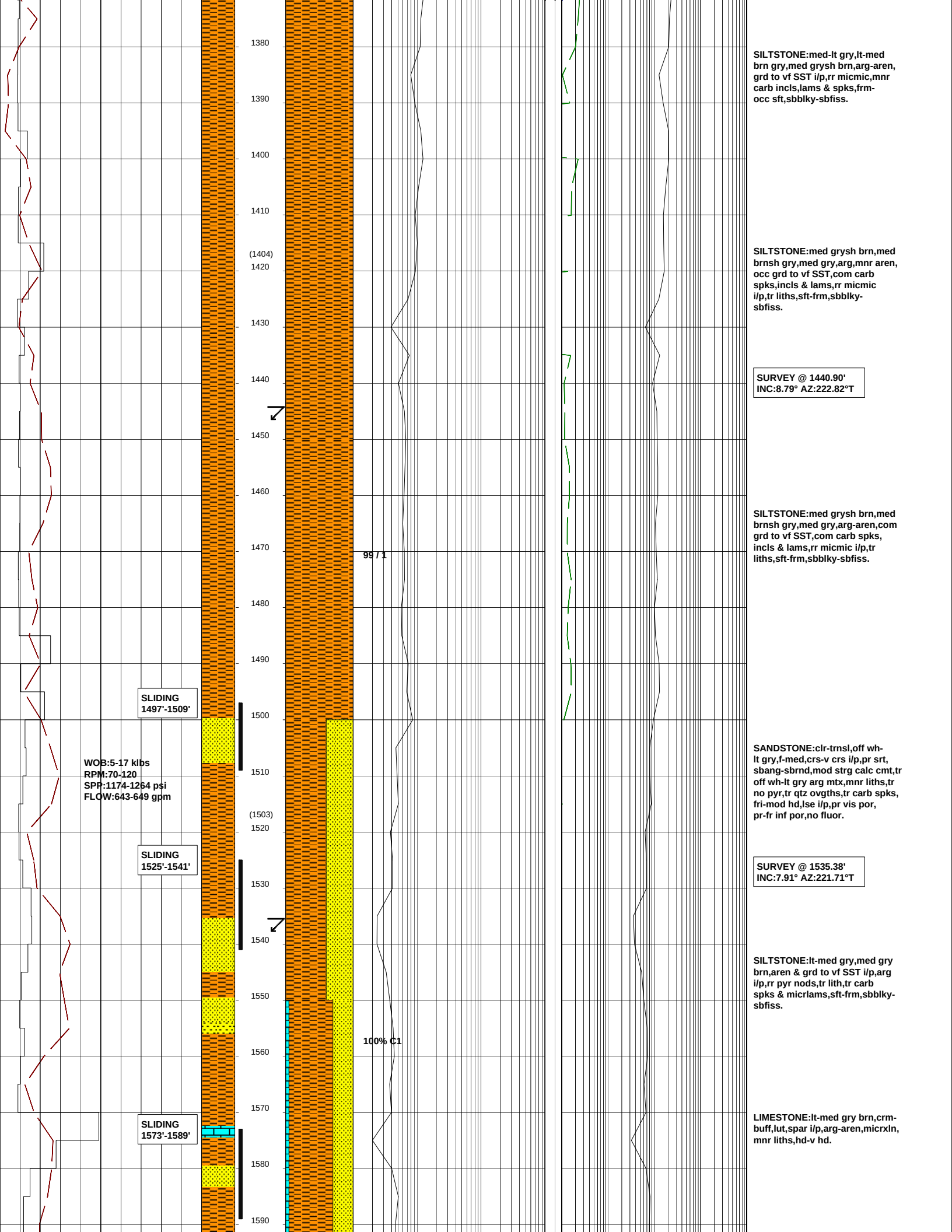
SILTSTONE:lt med gry,lt-med
grnsh gry,lt brnsh gry,arg-com
aren,occ grd to vf SST i/p,
occ liths,tr carb spks & flks,
tr disse pyr,frm-sft,
sbblky-sbfiss.

SURVEY @ 910.32'
INC:9.23° AZ:230.78°T

SANDSTONE:off wh-lt gry,vf f







SILTSTONE:med-lt gry,lt-med
brn gry,med grysh brn,arg-aren,
grd to vf SST i/p,rr micmic,mnr
carb incls,lams & spks,frm-
occ sft,sbblky-sbfiss.

SILTSTONE:med grysh brn,med
brnsh gry,med gry,arg,mnr aren,
occ grd to vf SST,com carb
spks,incls & lams,rr micmic
i/p,tr liths,sft-frm,sbblky-
sbfiss.

SURVEY @ 1440.90'
INC:8.79° AZ:222.82°T

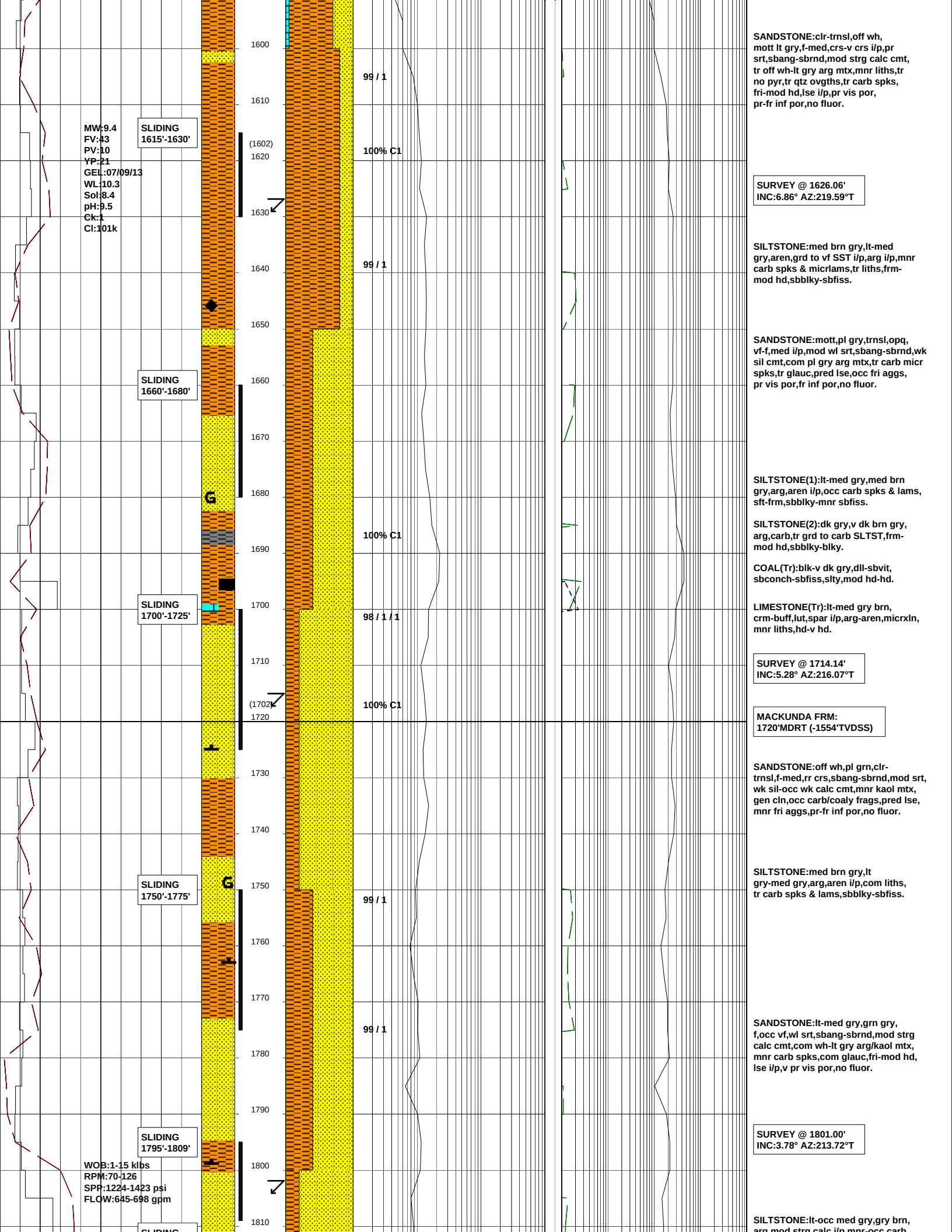
SILTSTONE:med grysh brn,med
brnsh gry,med gry,arg-aren,com
grd to vf SST,com carb spks,
incls & lams,rr micmic i/p,tr
liths,sft-frm,sbblky-sbfiss.

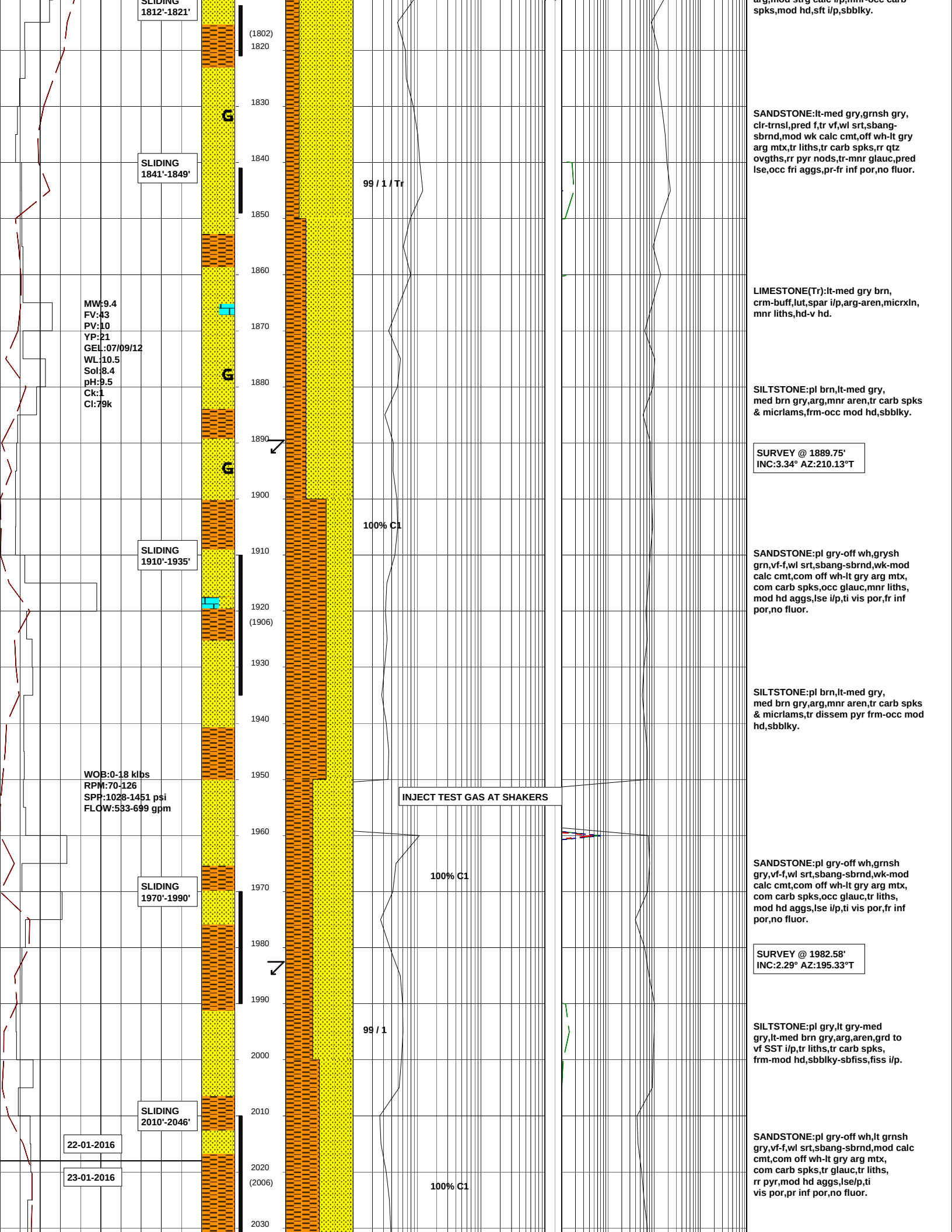
SANDSTONE:clr-trnsl,off wh-
lt gry,f-med,crs-v crs i/p,pr srt,
sbang-sbrnd,mod strg calc cmt,tr
off wh-lt gry arg mtx,mnr liths,tr
no pyr,tr qtz ovghts,tr carb spks,
fri-mod hd,lse i/p,pr vis por,
pr-fr inf por,no fluor.

SURVEY @ 1535.38'
INC:7.91° AZ:221.71°T

SILTSTONE:lt-med gry,med gry
brn,aren & grd to vf SST i/p,arg
i/p,rr pyr nods,tr lith,tr carb
spks & micrlams,sft-frm,sbblky-
sbfiss.

LIMESTONE:lt-med gry brn,crm-
buff,lut,spar i/p,arg-aren,micrxln,
mnr liths,hd-v hd.





Ck:1
Cl:153k

WOB:9-12 klbs
RPM:78-118
SPR:1380-1573 psi
FLOW:605-648 gpm

2260

2270

2280

2290

2300

2310

2320

(2311)
2330

2340

2350

2360

2370

2380

2390

2400

2410

2420

(2410)
2430

2440

2450

2460

2470

98 / 2

98 / 2

98 / 2 / Tr

96 / 2 / 1 / 1

SILTSTONE:med gry,occ pl gry,
pl brnsh gry,pred v arg,occ vf aren,
com carb spks-occ lams,tr
micmic & lithics,v sft-v frm,
sbbiky-blky

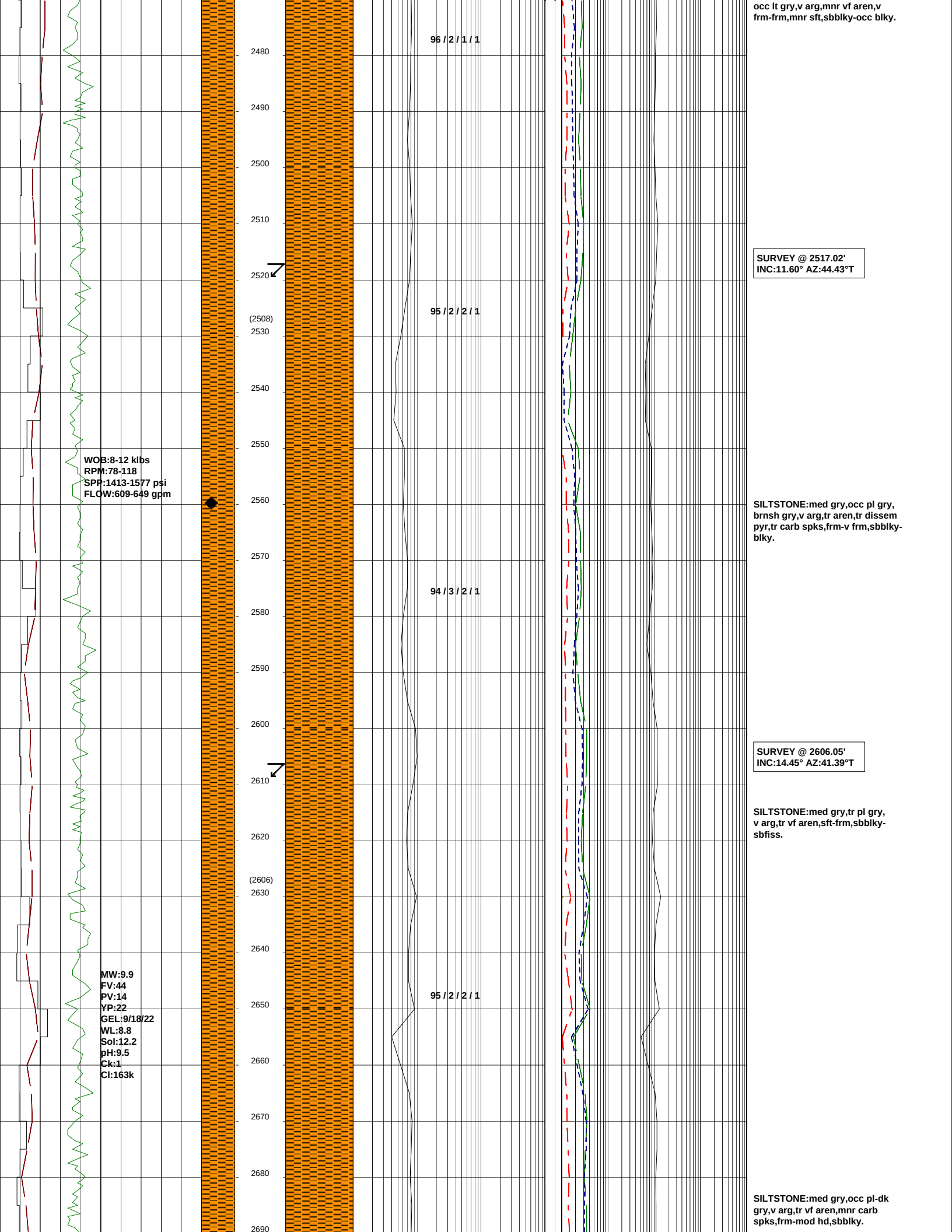
SILTSTONE:med-dk gry,brn gry,
occ lt gry,v arg,mnr vf aren,v
frm-frm,mnr sft,sbbiky-occ blky.

SURVEY @ 2338.83'
INC:6.01° AZ:58.64°T

SANDSTONE:off wh-wh,pl gry,
vf,grd to aren SLTST,sbang,wl
srt,frm med calc cmt,mnr kaol-
sity mtz,mod hd-fri aggs,v pr
vis por,no fluor.

SURVEY @ 2427.70'
INC:8.98° AZ:50.03°T

SILTSTONE:med-dk gry,brn gry,



occ lt gry,v arg,mnr vf aren,v
frm-frm,mnr sft,sbblky-occ blk.

SURVEY @ 2517.02'
INC:11.60° AZ:44.43°T

SILTSTONE: med gry, occ pl gry,
brnsh gry, v arg, tr aren, tr dissem
pyr, tr carb spks, frm-v frm, sbblky-
blk.

SURVEY @ 2606.05'
INC:14.45° AZ:41.39°T

SILTSTONE: med gry, tr pl gry,
v arg, tr vf aren, sft-frm, sbblky-
sbfiss.

SILTSTONE: med gry, occ pl-dk
gry, v arg, tr vf aren, mntr carb
spks, frm-mod hd, sbblky.

96 / 2 / 1 / 1

95 / 2 / 2 / 1

94 / 3 / 2 / 1

95 / 2 / 2 / 1

WOB: 8-12 klbs
RPM: 78-118
SPP: 1413-1577 psi
FLOW: 609-649 gpm

MW: 9.9
FV: 44
PV: 14
YP: 22
GEL: 9/18/22
WL: 8.8
Sol: 12.2
pH: 9.5
Ck: 1
CI: 163k

WOB:5-10 klbs
RPM:79-129
SPR:1421-1567 psi
FLOW:605-640 gpm

SURVEY @ 2694.15'
INC:14.78° AZ:42.39°T

92 / 3 / 3 / 2

SILTSTONE:med gry,occ pl-dk
gry,v arg,tr vf aren,mnr carb
spks,tr pyr,frm-mod hd,stkylp,
sbbilky.

SURVEY @ 2785.31'
INC:17.61° AZ:42.19°T

93 / 3 / 3 / 1

SILTSTONE:med gry,occ pl-dk
gry,v arg,tr vf aren,tr grd to vf
SST,mnr carb spks,tr pyr,frm-
mod hd,sbbilky.

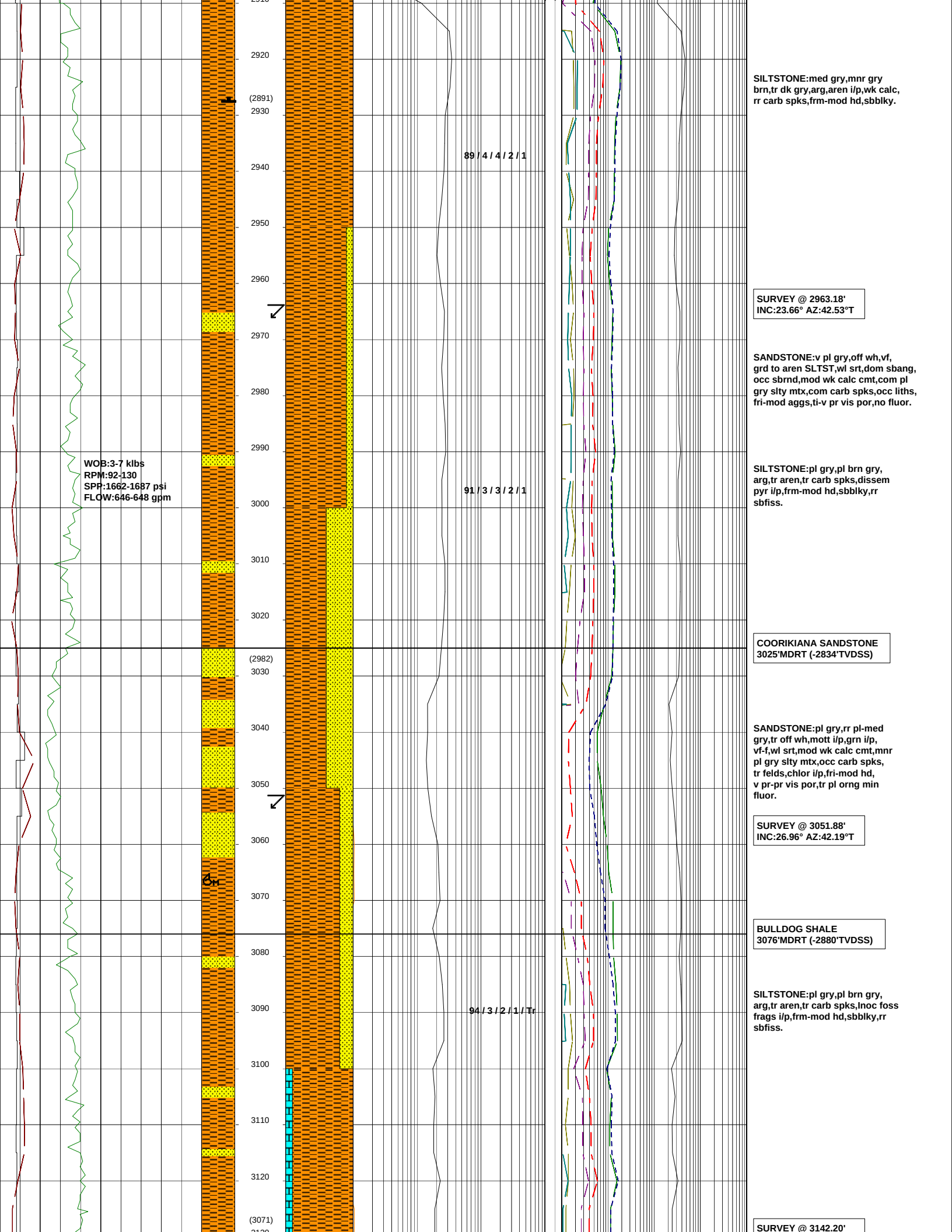
93 / 3 / 3 / 1

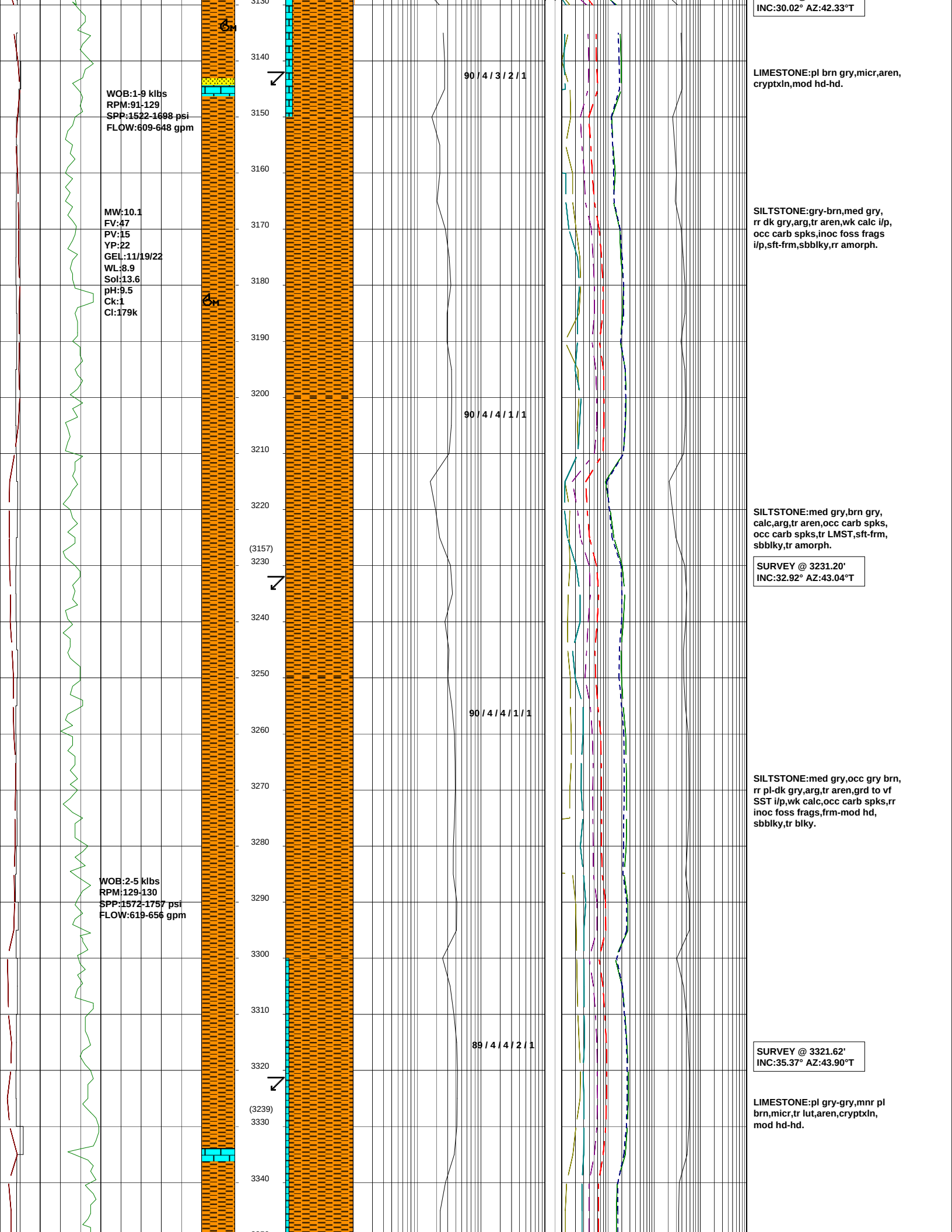
SILTSTONE:med gry,mnr gry brn,
tr dk gry,arg,wk calc,rr carb spks,
frm-mod hd,sbbilky.

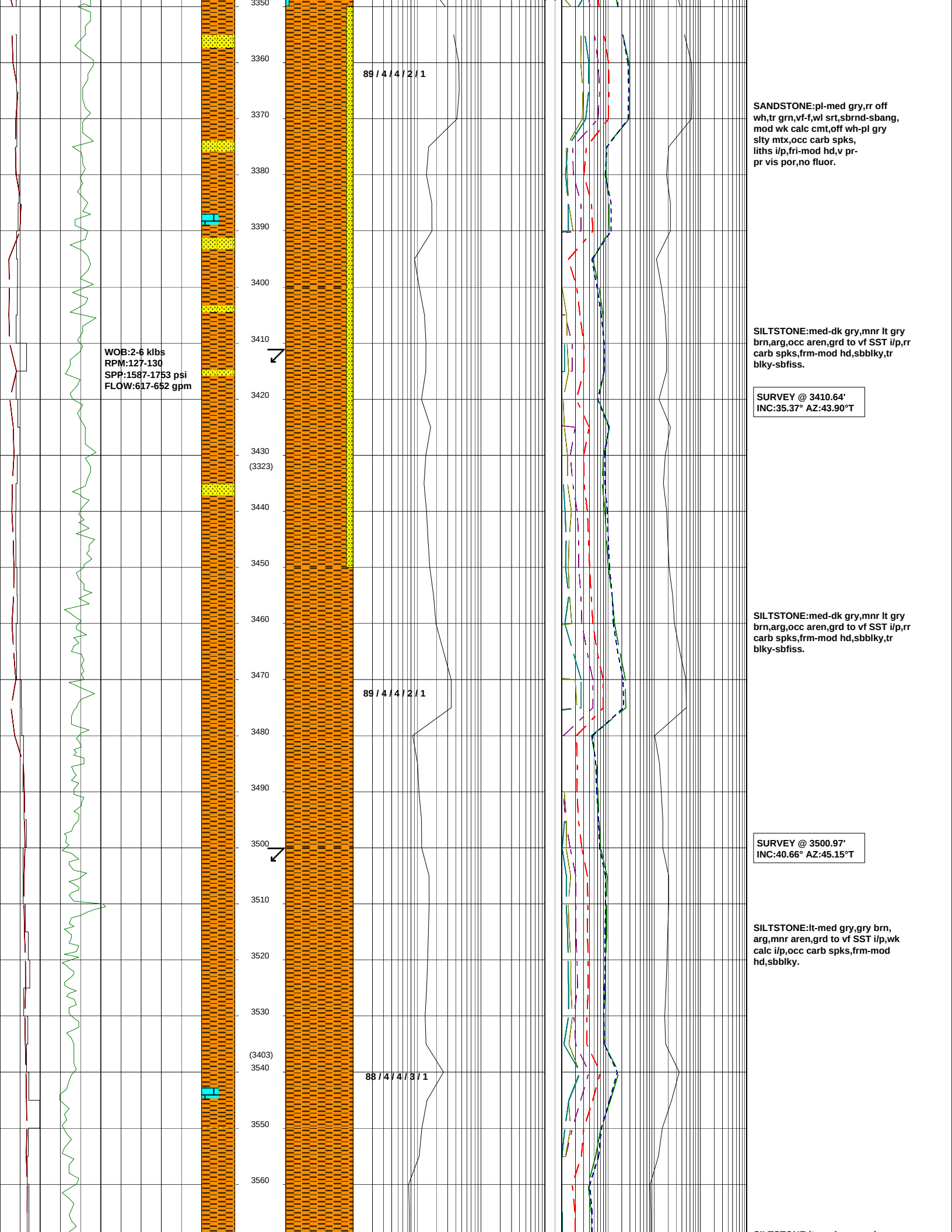
SURVEY @ 2873.66'
INC:20.60° AZ:42.47°T

90 / 4 / 3 / 2 / 1

WOB:5-8 klbs
RPM:117-129
SPR:1432-1676 psi
FLOW:606-646 gpm







SURVEY @ 3590.37'
INC:43.84° AZ:45.09°T

SILTSTONE:lt-med gry,gry brn,
arg,mnr aren,grd to vf SST i/p,wk
calc i/p,occ carb spks,frm-mod hd,
sbbly.

SILTSTONE: med gry, gry brn, arg, occ aren, grd to vf SST i/p, rr carb spks, inoc foss frags i/p, tr dissem pyr, sft-mod hd, sbbiky-tr sbfiss.

SURVEY @ 3679.47'
INC:47.23° AZ:45.39°T

SANDSTONE:off wh,pl gry,rr dk gry,vf-f,wl srt,sbrnd,mod calc cmt,occ pl gry slty mtx,com carb spks,fri-mod hd,v pr vis por,no fluor.

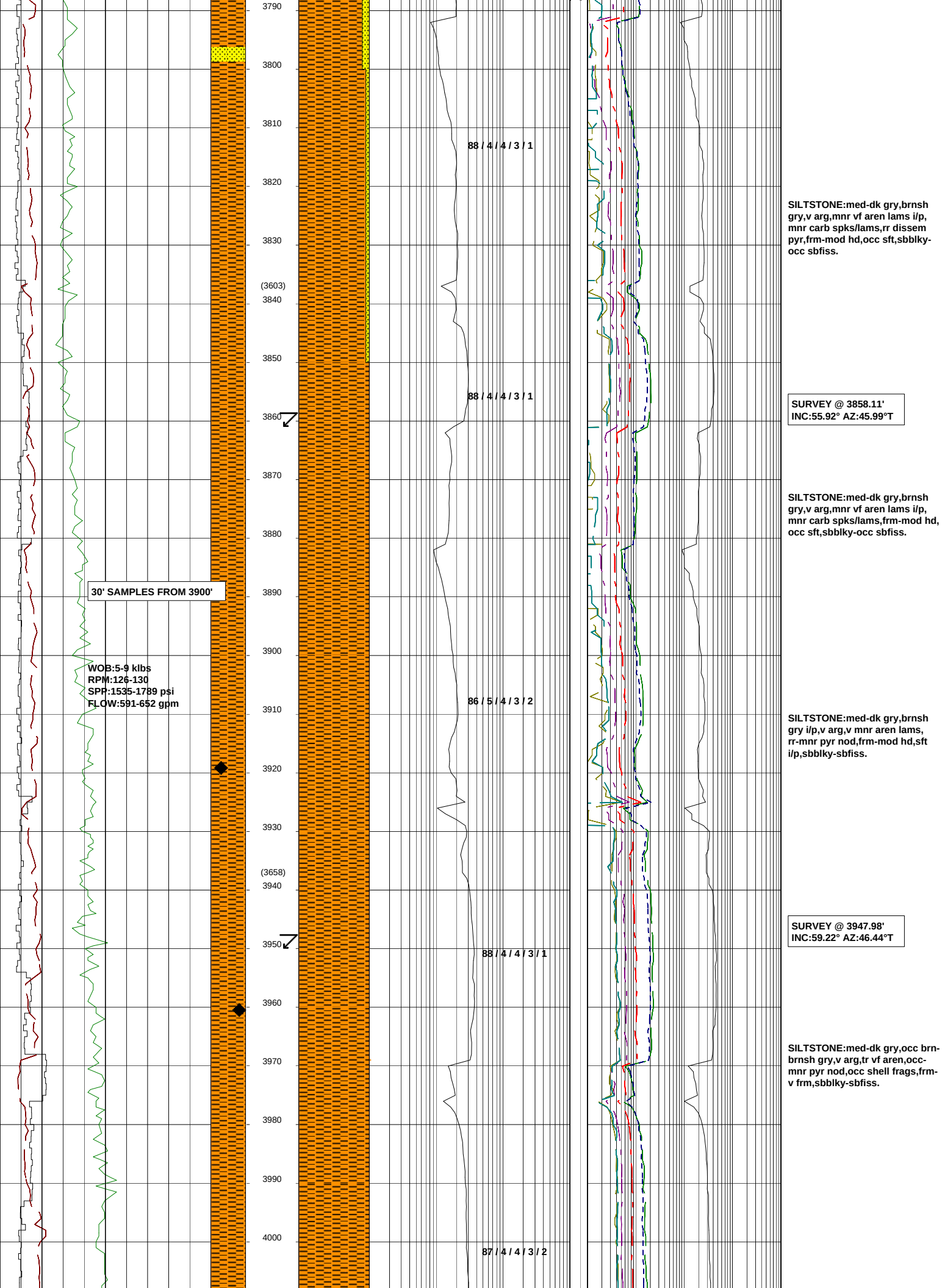
SURVEY @ 3768.3'
INC:51.84° AZ:45.77°T

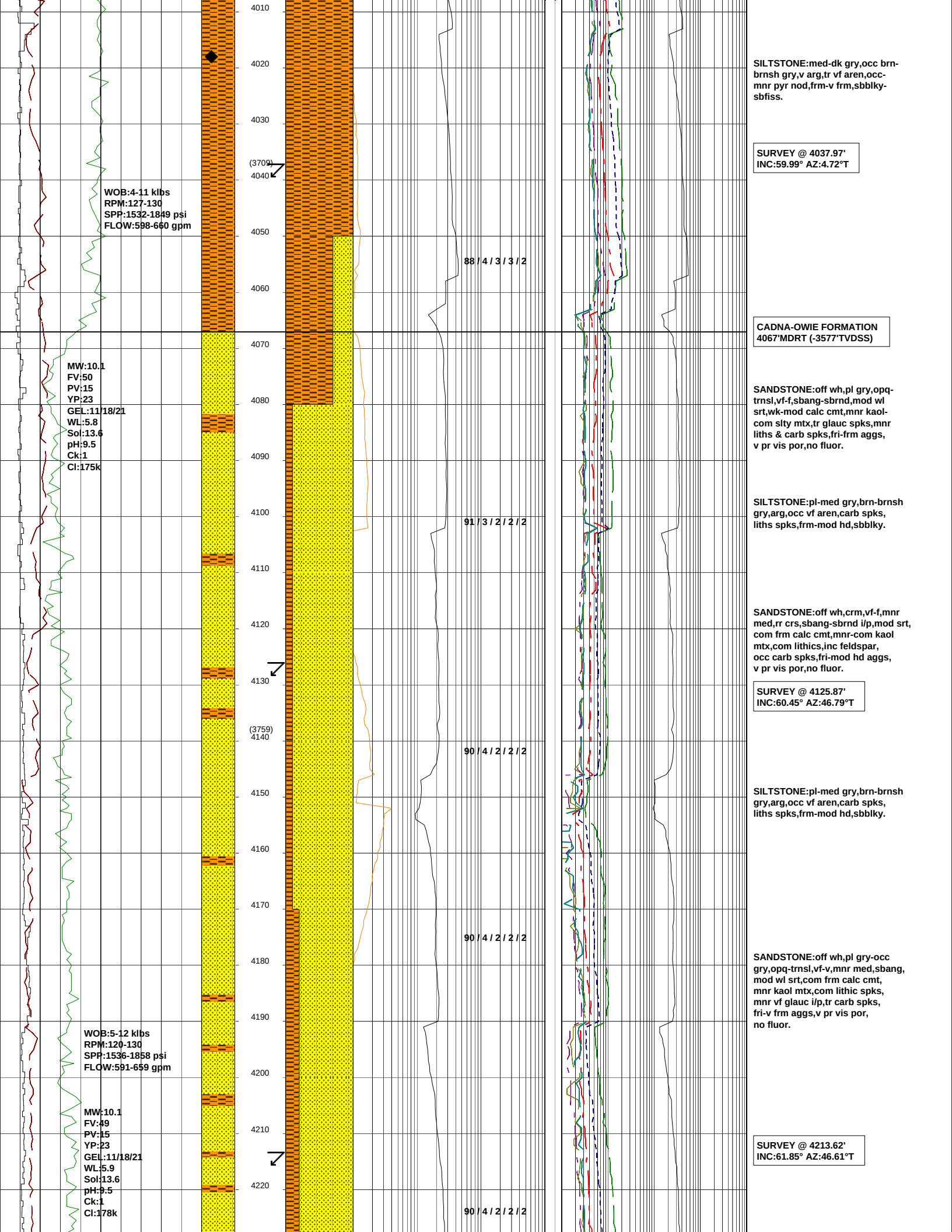
SANDSTONE:off wh-pl gry,opq-trnsl,vf-occ f,shang,mod wl srt,wk-frn calc cmt,mnr kaol-slty mtz,mnr carb,glaucl,liths,fri-mod hd aggs,v pr-ti vis por,no fluor.

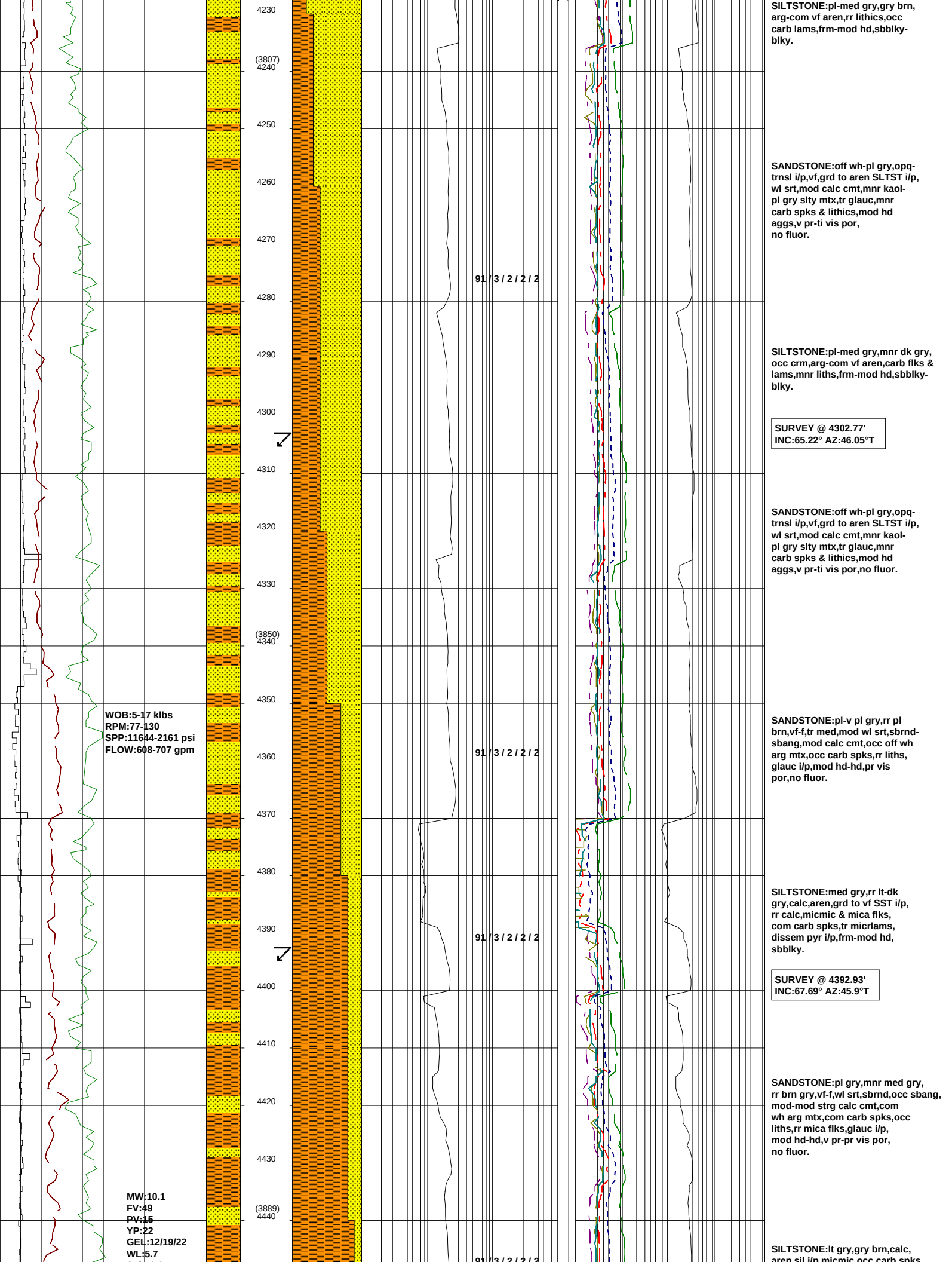
1' DATA BASE FROM 3600'

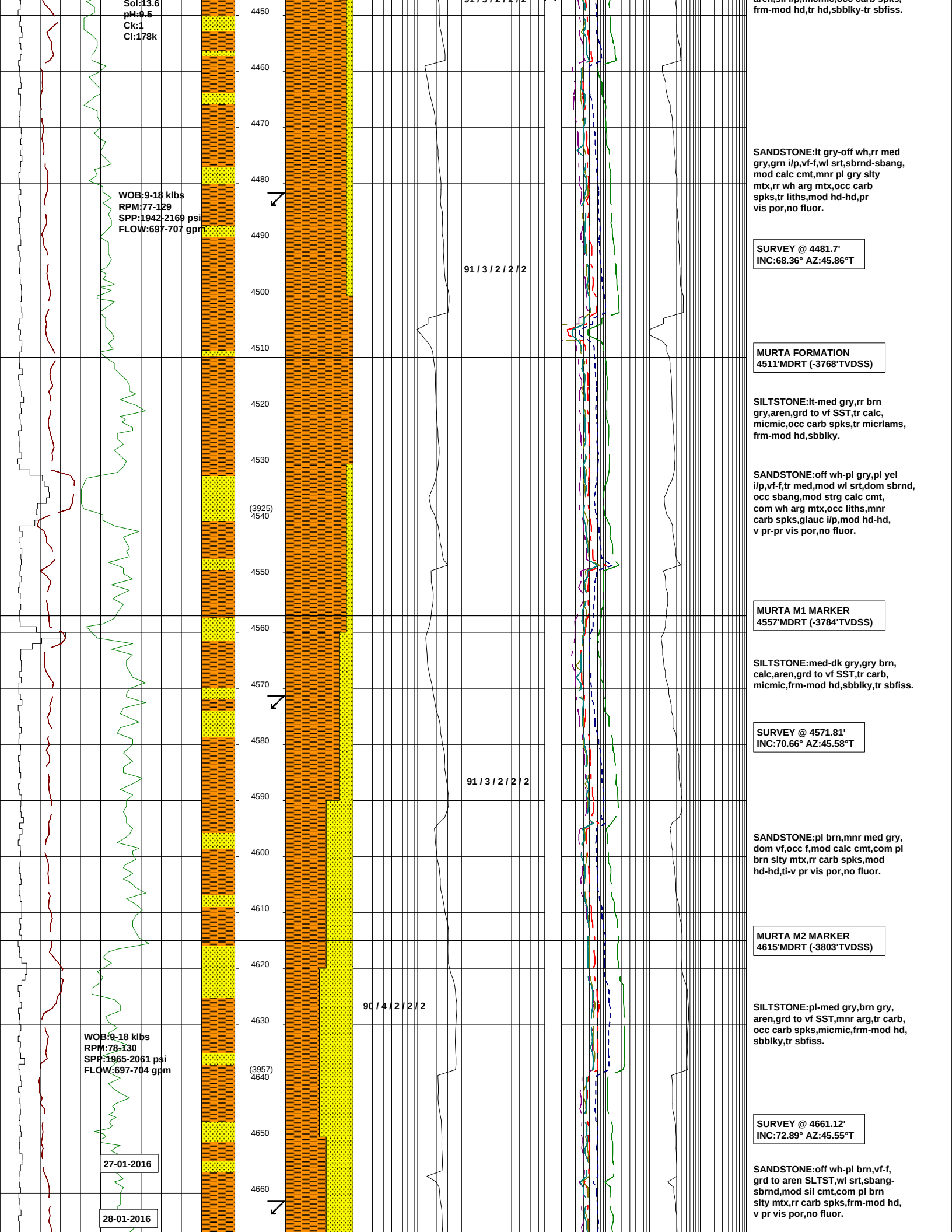
WOB:5-9 klbs
RPM:107-129
SPP:1555-1837 psi
FLOW:605-667 gpm

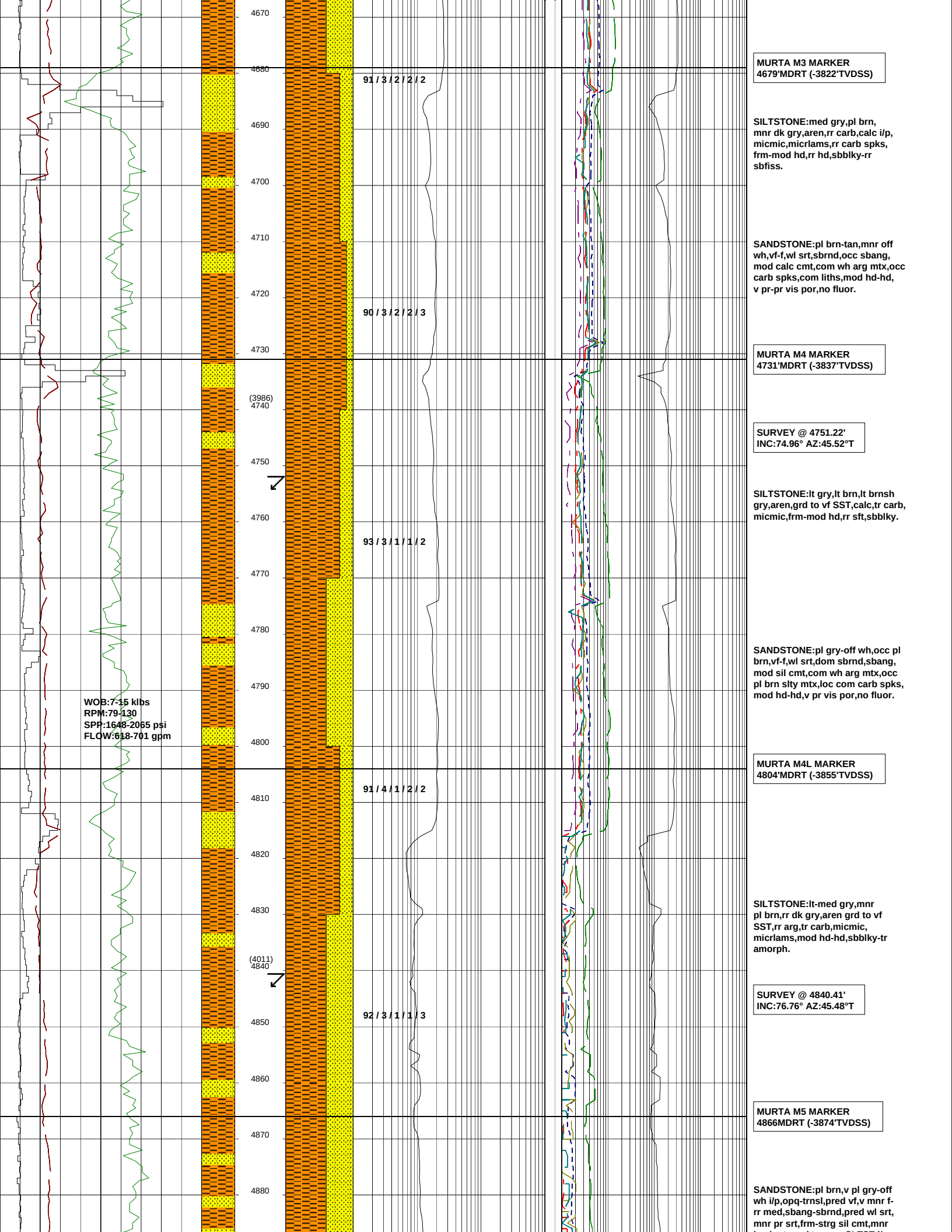
WOB:4-11 klbs	
RPM:80-129	
SPP:1555-1829 psi	
FLOW:605-662 gpm	











MURTA M3 MARKER
4679'MDRT (-3822'TVDSS)

SILTSTONE: med gry, pl brn,
mnr dk gry, aren, rr carb, calc i/p,
micmic, micrlams, rr carb spks,
frm-mod hd, rr hd, sbblky-rr
sbfiss.

SANDSTONE: pl brn-tan, mnr off
wh, vf-f, wl srt, sbnd, occ sbang,
mod calc cmt, com wh arg mtx, occ
carb spks, com liths, mod hd-hd,
v pr-pr vis por, no fluor.

MURTA M4 MARKER
4731'MDRT (-3837'TVDSS)

SURVEY @ 4751.22'
INC: 74.96° AZ: 45.52°T

SILTSTONE: lt gry, lt brn, lt brnsh
gry, aren, grd to vf SST, calc, tr carb,
micmic, frm-mod hd, rr sft, sbblky.

SANDSTONE: pl gry-off wh, occ pl
brn, vf-f, wl srt, dom sbnd, sbang,
mod sil cmt, com wh arg mtx, occ
pl brn slty mtx, loc com carb spks,
mod hd-hd, v pr vis por, no fluor.

MURTA M4L MARKER
4804'MDRT (-3855'TVDSS)

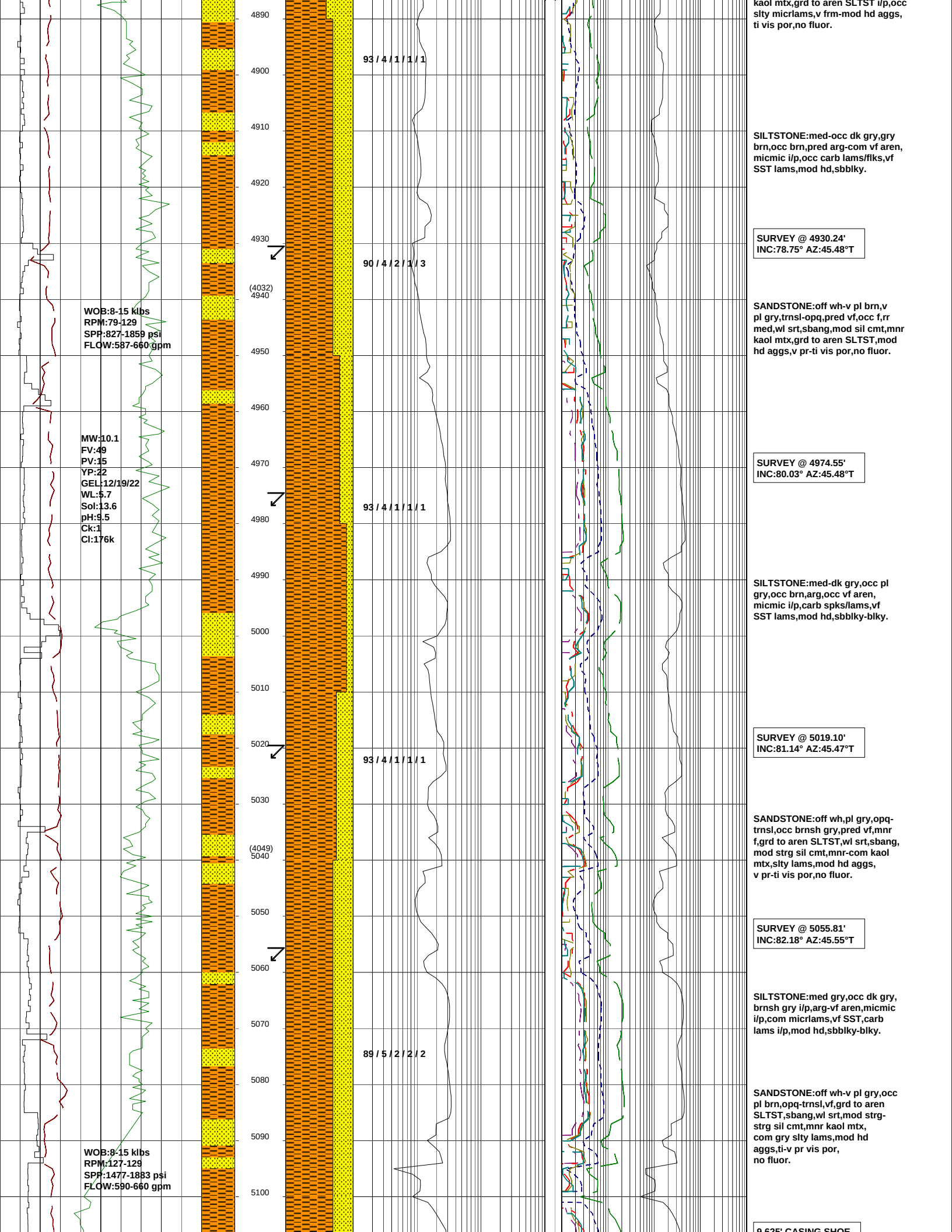
SILTSTONE: lt-med gry, mnr
pl brn, rr dk gry, aren grd to vf
SST, rr arg, tr carb, micmic,
micrlams, mod hd-hd, sbblky-tr
amorph.

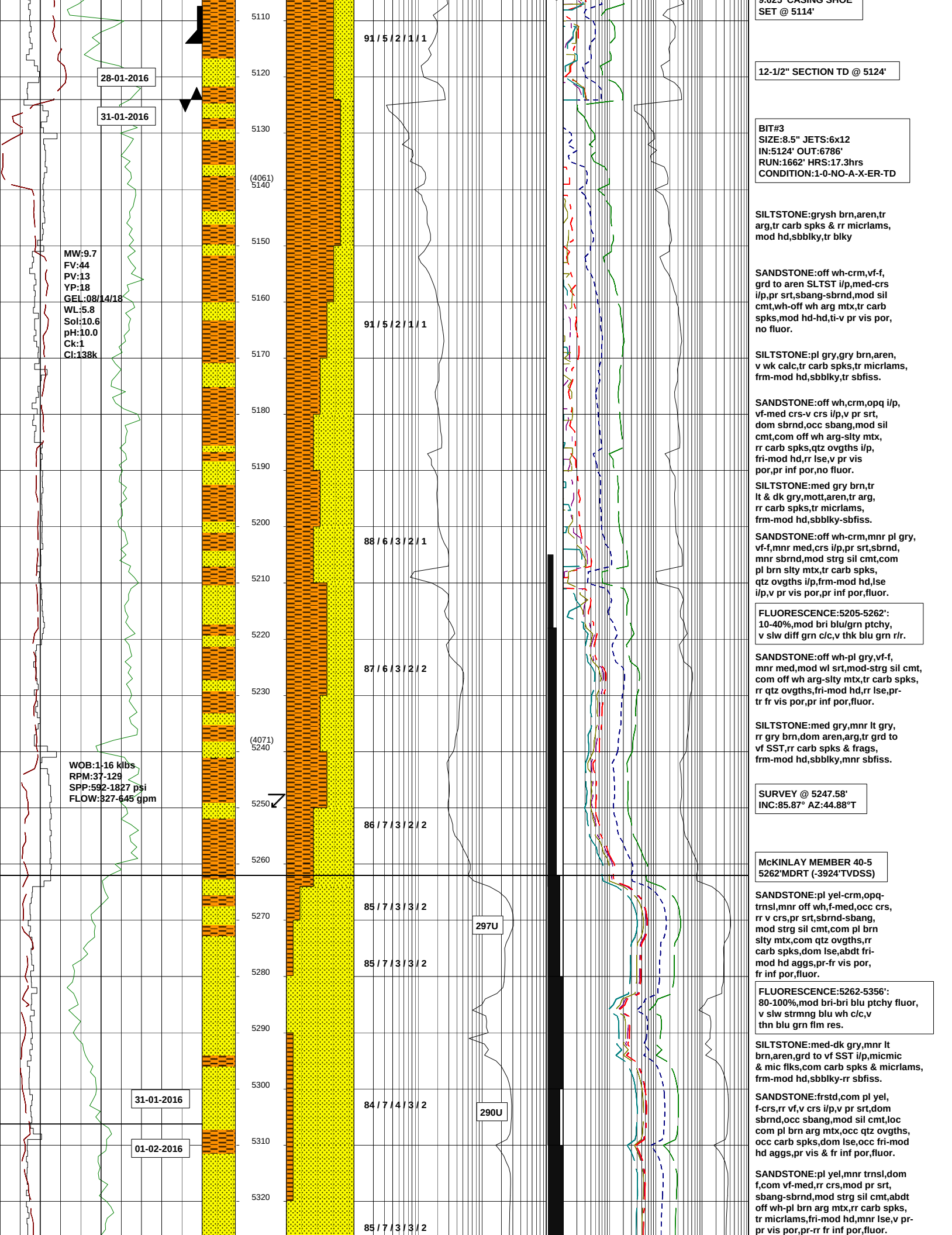
SURVEY @ 4840.41'
INC: 76.76° AZ: 45.48°T

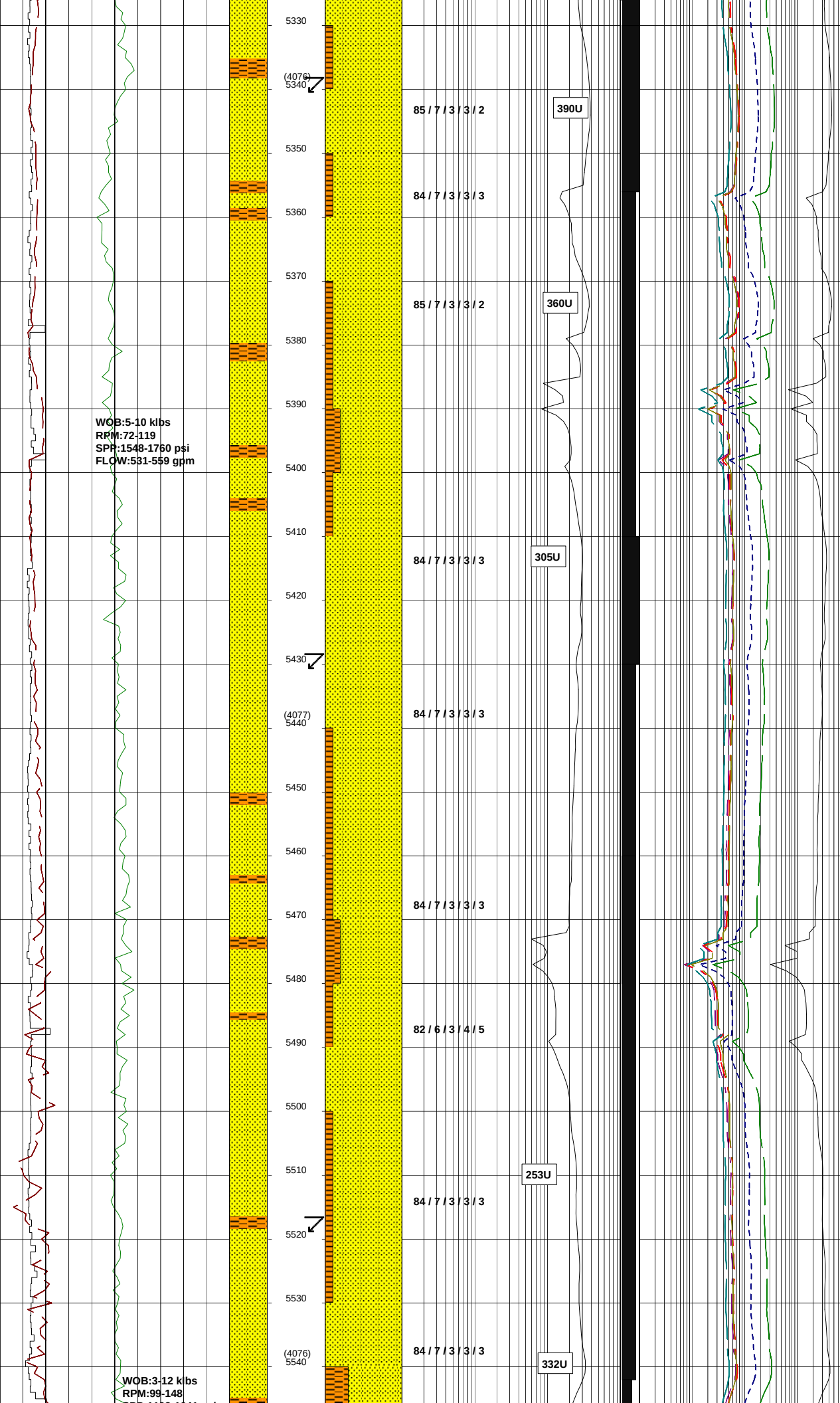
MURTA M5 MARKER
4866MDRT (-3874'TVDSS)

SANDSTONE: pl brn, v pl gry-off
wh i/p, opq-trnsi, pred vf, v mnr f-
rr med, sbang-sbnd, pred wl srt,
mnr pr srt, frm-strg sil cmt, mnr

WOB: 7-15 kips
RPM: 79-130
SPR: 1648-2065 psi
FLOW: 618-701 gpm







SURVEY @ 5338.68'
INC: 88.48° AZ: 44.0°T

SILTSTONE: med brn, aren, grd to vf SST, occ carb spks & lams, frm-mod hd, sbbkly.

SANDSTONE: pl brn-off wh, f-med, rr vf, mod wl srt, dom sbrnd, occ sbang, mod sil cmt, com pl brn arg mtz, rr carb spks, tr mic flks, fri-mod hd aggs, rr lse, pr-mnr fr vis & inf por, fluor.

FLUORESCENCE: 5356-5480':
60-90%, pl-mod bri blu grn evn, v slw bldng cut, mod thn blu flm res.

SANDSTONE: off wh-v pl brn, dom f, mnr vf-med, mod wl srt, sbang-sbrnd, mod sil cmt, mnr off wh arg mtz, rr carb spks, fri-mod hd, occ lse, pr-fr vis & inf por, fluor.

SILTSTONE: med brn, mnr gry brn, aren, occ micrlams, rr carb spks, frm-mod hd, sbbkly-rr sbfiss.

SANDSTONE: pl yel brn, rr off wh, f, occ vf-med, mod wl srt, sbang-sbrnd, mod sil cmt, mnr off wh arg mtz, rr carb spks & micrlams, fri-mod hd aggs, occ lse, pr vis por, pr-fr inf por, fluor.

SANDSTONE: pl yel brn, rr off wh, f, occ vf-med, mod wl srt, sbang-sbrnd, mod sil cmt, mnr off wh arg mtz, rr carb spks & micrlams, fri-mod hd aggs, occ lse, pr vis por, pr-fr inf por, fluor.

SURVEY @ 5428.33'
INC: 90.09° AZ: 43.74°T

SANDSTONE: pl yel brn, rr off wh, f, occ vf-med, mod wl srt, sbang-sbrnd, mod sil cmt, mnr off wh arg mtz, rr carb spks & micrlams, fri-mod hd aggs, occ lse, pr vis por, pr-fr inf por, fluor.

SILTSTONE: pl gry brn, tr dk brn, aren, carb micrlams i/p, frm-mod hd, sbbkly.

SANDSTONE: pl yel brn, rr off wh, f, occ vf-med, mod wl srt, sbang-sbrnd, mod sil cmt, mnr off wh arg mtz, rr carb spks & micrlams, fri-mod hd aggs, occ lse, pr vis por, pr-fr inf por, fluor.

SANDSTONE: pl yel brn, mnr off wh, vf-f, rr med, wl srt, sbang-sbrnd, mod sil cmt, com off wh arg-sltz mtz, rr carb spks, qtz ovgrhs i/p, fri-mod hd aggs, rr lse, v pr-pr vis por, pr inf por, fluor.

SANDSTONE: off wh, crm, dom f, occ vf-med, mod wl srt, sbrnd-sbang, mod sil cmt, com off wh arg mtz, occ carb spks, fri-mod hd aggs, com lse, pr vis & inf por, fluor.

SILTSTONE: pl gry-med brn, aren, grd to vf SST, occ micrlams, frm-mod hd, sbbkly.

SURVEY @ 5516.5'
INC: 90.06° AZ: 43.72°T

SANDSTONE: off wh, crm, dom f, occ vf-med, mod wl srt, sbrnd-sbang, mod sil cmt, com off wh arg mtz, occ carb spks, fri-mod hd aggs, com lse, pr vis & inf por, fluor.

SANDSTONE: pl yel brn, vf-f, wl srt, dom sbrnd, occ sbang, mod sil cmt, com off wh-pl brn arg mtz, rr carb spks, fri-mod hd, abdt lse, pr-fr vis & inf por, fluor.

FLUORESCENCE: 5542-5680':
50-90%, mod bri-bri blu/grn ptchy, v slw blu/wh strrng c/c, blu/wh thk r/r.

SILTSTONE: med gry brn, aren, grd to vf SST, occ carb spks & lams, frm-mod hd, sbbkly.

SPP:1192-1641 psi
FLOW:453-542 gpm

WOB:7-15 klbs
RPM:99-149
SPP:1296-1655 psi
FLOW:465-534 gpm

5550
5560
5570
5580
5590
5600
5610
5620
5630
(4075)
5640
5650
5660
5670
5680
5690
5700
5710
5720
5730
(4074)
5740
5750
5760

83 / 6 / 3 / 4 / 4

85 / 6 / 3 / 3 / 3

84 / 7 / 3 / 3 / 3

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84 / 7 / 3 / 3 / 3

82 / 6 / 4 / 4 / 4

81 / 7 / 4 / 4 / 4

83 / 7 / 4 / 3 / 3

84 / 7 / 3 / 3 / 3

82 / 7 / 4 / 4 / 3

370U

290U

263U

SILTSTONE:pl med gry brn,cc,grd
to vf SST,tr arg,occ carb spks &
micrlams,tr micmic,frm-mod hd,sbbiky.

SANDSTONE:pl yel-pl brn,vf-f,
wl srt,dom sbrnd,occ sbang,mod-
mod strg sil cmt,occ off wh-pl
gry arg-slty mtx,occ carb spks,
fri-mod hd,com lse,v pr vis por,
pr-mnr fr inf por,fluor.

SILTSTONE:pl-med brn,aren,mnr arg,
com carb spks,frm-mod hd,sbbiky.

SANDSTONE:off wh-v pl yel brn,
vf-f,wl srt,dom sbrnd,occ sbang,
mod strg sil cmt,com off wh arg
mtx,occ carb spks,fri-mod hd,com
lse,pr-fr vis & inf por,fluor.

SILTSTONE:med brn gry,mnr v pl
gry,dom aren,occ carb spks & loc
com micrlams,frm-mod hd,sbbiky.

SURVEY @ 5605.15'
INC:91.0° AZ:43.63°T

SANDSTONE:off wh-trnsl,v pl brn
i/p,vf-f,sbang,wl srt,wk-frm sil
cmt,mnr kaol mtx,tr liths spks,carb
spks,fri-frm aggs,v pr-pr vis por,fluor.

SANDSTONE:off wh-trnsl,v pl brn
i/p,vf-f,sbang,wl srt,wk-frm sil cmt,
mnr kaol mtx,tr liths spks,carb spks,
fri-frm aggs,v pr-pr vis por,fluor.

SANDSTONE:off wh-v pl brn,trnsl,
vf-occ f,sbang,wl srt,frm sil cmt,
mnr-com kaol mtx,occ liths/carb
spks,fri-frm aggs,v pr vis por,fluor.

SILTSTONE:pl-med gry,arg-vf
aren,micmic,carb spks,frm,sbbiky.

SANDSTONE:off wh-trnsl,v pl brn
i/p,vf,sbang,wl srt,wk-frm sil cmt,
mnr kaol mtx,tr carb spks,fri aggs,
v pr vis por,fluor.

FLUORESCENCE:5680-5840':
70-90%,mod bri-occ bri yel/grn,
evn-ptchy,slw diff c/c,mod-
thk blu/wh flm res.

SURVEY @ 5697.2'
INC:90.43° AZ:43.73°T

SILTSTONE(Tr):gry,vf aren,mnr arg,
micmic,mnr carb spks,frm-mod hd,
sbbiky.

SANDSTONE:off wh-trnsl,pl brn,
vf-pred f aggs,occ lse,mod-rr lse
crs,sbang-sbrnd,mod wl srt,frm sil
cmt,mnr kaol mtx,fri-v frm aggs,
mnr lse,v pr vis por,fluor.

